



AIRLINE FLIGHTS MANAGEMENT

MAKING TRAVEL EASY AND MEMORABLE

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AGENDA



MYSQL

- Data Collection
- Data Cleaning
- Transformation and Normalization

DashBoard

- Connection to Power BI
- Visualization

DataSet

Airline Dataset

Navigating the Skies: Exploring Insights from Synthetic Airline Data



<https://www.kaggle.com/datasets/iamsouravbanerjee/airline-dataset>

IMPORT DATA

Limit to 1000 rows

```

1 • create database airline;
2 • use airline;
3
4 • SELECT * FROM Airline;
5

```

	Passenger ID	First Name	Last Name	Gender	Age	Nationality	Airport Name	Airport Country Code	Country Name	Airport Continent	Continents	Departure Date	Arrival Airport
▶	ABVW1g	Edithe	Leggis	Female	62	Japan	Coldfoot Airport	US	United States	NAM	North America	6/28/2022	CXF
	jOXAX	Elwood	Catt	Male	62	Nicaragua	Kugluktuk Airport	CA	Canada	NAM	North America	12/26/2022	YCO
	CdUz2g	Darby	Felgate	Male	67	Russia	Grenoble-Isère Airport	FR	France	EU	Europe	1/18/2022	GNB
	BRS38v	Dominica	Pyle	Female	71	China	Ottawa / Gatineau Airport	CA	Canada	NAM	North America	9/16/2022	YND
	9kvTLo	Bay	Pencost	Male	21	China	Gillespie Field	US	United States	NAM	North America	2/25/2022	SEE
	nMJKVh	Lora	Durbann	Female	55	Brazil	Coronel Horácio de Mattos Airport	BR	Brazil	SAM	South America	06-10-2022	LEC
	8SPFPE	Rand	Bram	Male	73	Ivory Coast	Duxford Aerodrome	GB	United Kingdom	EU	Europe	10/30/2022	QFO
	pqbBY	Perceval	Dalosso	Male	36	Vietnam	Maestro Wilson Fonseca Airport	BR	Brazil	SAM	South America	04-07-2022	STM
	QHAs2R	Aleda	Pigram	Female	35	Palestinian Territory	Venice Marco Polo Airport	IT	Italy	EU	Europe	8/20/2022	VCE
	Jymudz	Burke	Schusti	Male	13	Thailand	Vermilion Airport	CA	Canada	NAM	North America	04-06-2022	YVG
	ZP41qZ	Porty	Jori	Male	39	Tunisia	Nuevo Casas Grandes Airport	MX	Mexico	NAM	North America	5/27/2022	NGG
	sBF524	Briant	De La Haye	Male	71	Russia	Ruben Cantu Airport	PA	Panama	NAM	North America	02-06-2022	SYP
	PlwJZT	Kale	Scoble	Female	47	Sweden	Loralai Airport	PK	Pakistan	AS	Asia	3/19/2022	LRG
	iJ75x3	Catriona	Beaumont	Female	77	Russia	Cudal Airport	AU	Australia	OC	Oceania	3/24/2022	CUJ
	GUTA6R	Amberly	Handling	Female	32	China	Farmington Regional Airport	US	United States	NAM	North America	03-07-2022	FAM
	8qa80a	Dyna	De Vere - ...	Female	22	China	Oudtshoorn Airport	ZA	South Africa	AF	Africa	7/18/2022	OUH

Airline 1 x

Output

#	Time	Action	Message
✓ 19	19:23:06	DEALLOCATE PREPARE stmt	OK
✓ 20	19:23:14	use airline	0 row(s) affected
✓ 21	19:23:20	SELECT * FROM Airline LIMIT 0, 1000	126 row(s) returned

Data Cleaning

```
ALTER TABLE Airline  
DROP COLUMN `Airport Continent`,  
DROP COLUMN `Airport Country Code`;
```


Data Cleaning

```
SELECT *  
FROM Airline  
WHERE  
    `First Name` IS NULL OR  
    `Last Name` IS NULL OR  
    Gender IS NULL OR  
    Age IS NULL OR  
    Nationality IS NULL OR  
    `Airport Name` IS NULL OR  
    `Country Name` IS NULL OR  
    `Continents` IS NULL OR  
    `Arrival Airport` IS NULL OR  
    `Departure Date` IS NULL OR  
    `Flight Status` IS NULL OR  
    `Pilot Name` IS NULL;
```

Data Normalization and Transformation

1- Passenger Table

```
CREATE TABLE Passengers(Passenger_ID INT AUTO_INCREMENT PRIMARY KEY) AS
SELECT DISTINCT
    CONCAT(`First Name`, ' ', `Last Name`) AS Passenger_Name,
    Gender,
    Age,
    Nationality,
    `Airport Name` AS Airport_Name
FROM Airline;

ALTER TABLE Passengers
ADD COLUMN Number_of_Flights INT DEFAULT 0;
UPDATE Passengers
SET Number_of_Flights = ROUND(RAND() * 8 + 2);

ALTER TABLE Passengers
ADD COLUMN Travel_Class VARCHAR(20);
UPDATE Passengers
SET Travel_Class =
    CASE
        WHEN Passenger_ID % 3 = 0 THEN 'Economy'
        WHEN Passenger_ID % 3 = 1 THEN 'Business'
        ELSE 'First'
    END;
```

Data Normalization and Transformation

```
1 ADD COLUMN Ticket_Price DECIMAL(10,2);
4 • UPDATE Passengers
5 SET Ticket_Price =
6 CASE
7 WHEN Travel_Class = 'Economy' THEN 1000.00
8 WHEN Travel_Class = 'Business' THEN 1500.00
9 WHEN Travel_Class = 'First' THEN 2000.00
10 ELSE 1200.00
11 END;
12
13 • SELECT * FROM Passengers;
14
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

Passenger_ID	Passenger_Name	Gender	Age	Nationality	Airport_Name	Number_of_Flights	Travel_Class	Ticket_Price
1	Edith Leggis	Female	62	Japan	Coldfoot Airport	3	Business	1500.00
2	Elwood Catt	Male	62	Nicaragua	Kugluktuk Airport	8	First	2000.00
3	Darby Felgate	Male	67	Russia	Grenoble-Isère Airport	8	Economy	1000.00
4	Dominica Pyle	Female	71	China	Ottawa / Gatineau Airport	4	Business	1500.00
5	Bay Percost	Male	21	China	Gillespie Field	9	First	2000.00
6	Lora Durbann	Female	55	Brazil	Coronel Horácio de Mattos Airport	2	Economy	1000.00
7	Rand Bram	Male	73	Ivory Coast	Duxford Aerodrome	4	Business	1500.00
8	Perceval Dalosso	Male	36	Vietnam	Maestro Wilson Fonseca Airport	6	First	2000.00
9	Aleda Pigram	Female	35	Palestinian Territory	Venice Marco Polo Airport	5	Economy	1000.00
10	Boris Khrushch	Male	17	Thailand	Varadero Airport	6	Business	1500.00

Passengers 3 x | Apply | Rev

Data Normalization and Transformation

2- Airport Table

```
CREATE TABLE Airport(Airport_ID INT AUTO_INCREMENT PRIMARY KEY) AS
SELECT DISTINCT
    `Airport Name` AS Airport_Name,
    `Country Name` AS Country_Name,
    `Arrival Airport` AS Arrival_Airport,
    Continents
FROM Airline;

ALTER TABLE Airport
ADD COLUMN Total_Revenue DECIMAL(12,2);

UPDATE Airport ar
SET Total_Revenue = (
    SELECT SUM(p.Ticket_Price * p.Number_of_Flights)
    FROM Passengers p
    WHERE p.Airport_Name = ar.Airport_Name
);
```

Data Normalization and Transformation

95

96 • `SELECT * FROM Airport;`

97

Result Grid

	Airport_ID	Airport_Name	Country_Name	Arrival_Airport	Continents	Total_Revenue
▶	1	Coldfoot Airport	United States	CXF	North America	4500.00
	2	Kugluktuk Airport	Canada	YCO	North America	16000.00
	3	Grenoble-Isère Airport	France	GNB	Europe	8000.00
	4	Ottawa / Gatineau Airport	Canada	YND	North America	6000.00
	5	Gillespie Field	United States	SEE	North America	18000.00
	6	Coronel Horácio de Mattos Airport	Brazil	LEC	South America	2000.00
	7	Duxford Aerodrome	United Kingdom	QFO	Europe	6000.00
	8	Maestro Wilson Fonseca Airport	Brazil	STM	South America	12000.00
	9	Venice Marco Polo Airport	Italy	VCE	Europe	5000.00
	10	Vermilion Airport	Canada	YVG	North America	7500.00
	11	Nuevo Casas Grandes Airport	Mexico	NCG	North America	16000.00

Airport 4 x App

Output

Action Output

#	Time	Action	Message
35	19:44:19	ALTER TABLE Airport ADD COLUMN Total_Revenue DECIMAL(12,2)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
36	19:44:19	UPDATE Airport ar SET Total_Revenue = (SELECT SUM(p.Ticket_Price * p.Number_of_Flights) FROM ...	125 row(s) affected Rows matched: 125 Changed: 125 Warnings: 0
37	19:44:24	SELECT * FROM Airport LIMIT 0, 1000	125 row(s) returned

Data Normalization and Transformation

3- Flight_Status Table

```
104 • CREATE TABLE Flight_Status(FlightStatus_ID INT AUTO_INCREMENT PRIMARY KEY) AS
105     SELECT DISTINCT `Flight Status` AS Flight_Status
106     FROM Airline;
107
108 • ALTER TABLE Flight_Status
109     ADD COLUMN Total_Number INT DEFAULT 0;
110
111 • UPDATE Flight_Status fs
112     JOIN (
113         SELECT `Flight Status` AS Flight_Status, COUNT(*) AS Status_Count
114         FROM Airline
115         GROUP BY `Flight Status`
116     ) AS counts ON fs.Flight_Status = counts.Flight_Status
117     SET fs.Total_Number = counts.Status_Count;
```

Data Normalization and Transformation

```
9 • SELECT * FROM Flight_Status;
```

```
0
```

```
1
```

ult Grid



Filter Rows:

Edit:



FlightStatus_ID	Flight_Status	Total_Number
1	On Time	43
2	Delayed	40
3	Cancelled	43

Data Normalization and Transformation

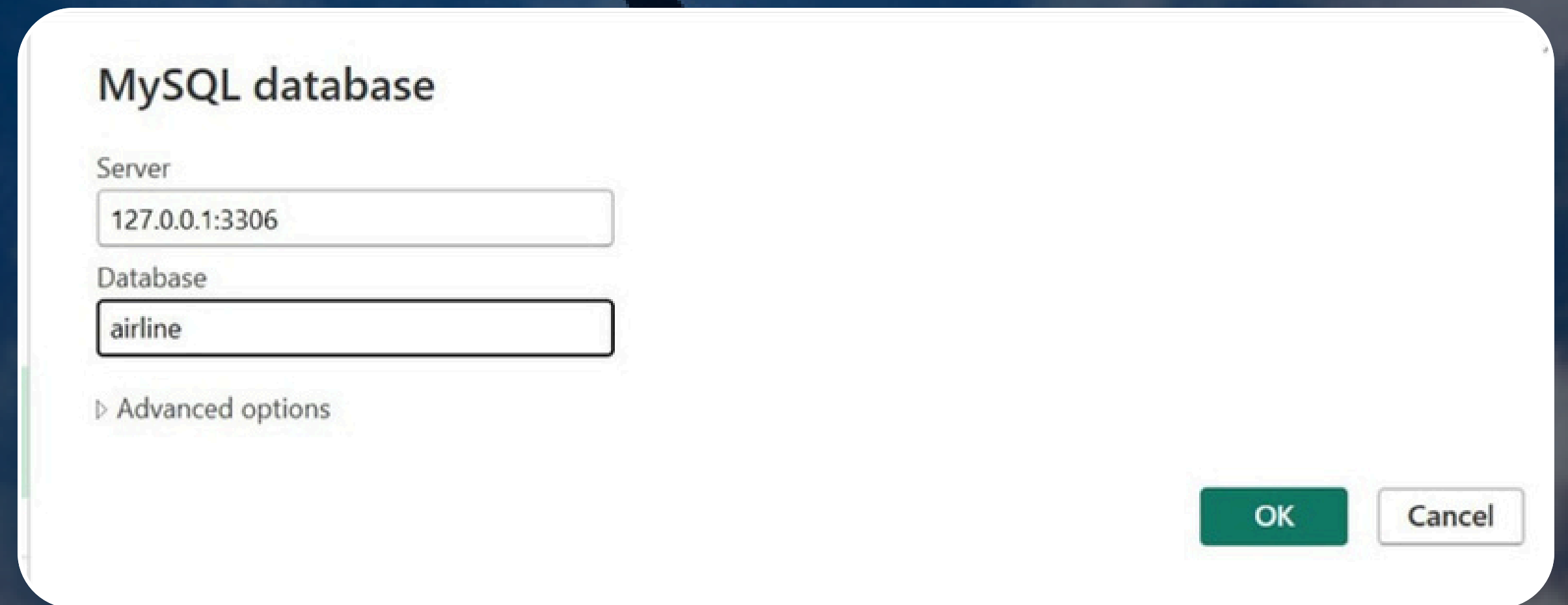
4- Pilot Table

```
6 • CREATE TABLE Pilot(Pilot_ID INT AUTO_INCREMENT PRIMARY KEY) AS
7   SELECT DISTINCT `Pilot Name` AS Pilot_Name
8   FROM Airline;
9
10 • ALTER TABLE Pilot
11   ADD COLUMN Salary DECIMAL(10,2);
12
13 • UPDATE pilot
14   SET salary = ROUND(RAND() * 5000 + 10000, 2);
15
16 • SELECT * FROM Pilot;
```

Pilot_ID	Pilot_Name	Salary
1	Fransisco Hazeldine	12056.89
2	Marla Parsonage	14635.59
3	Rhonda Amber	12007.30

Connection Code in Power BI

- Sever Name
- DataBase
- Port
- UserName
- Password



MySQL database

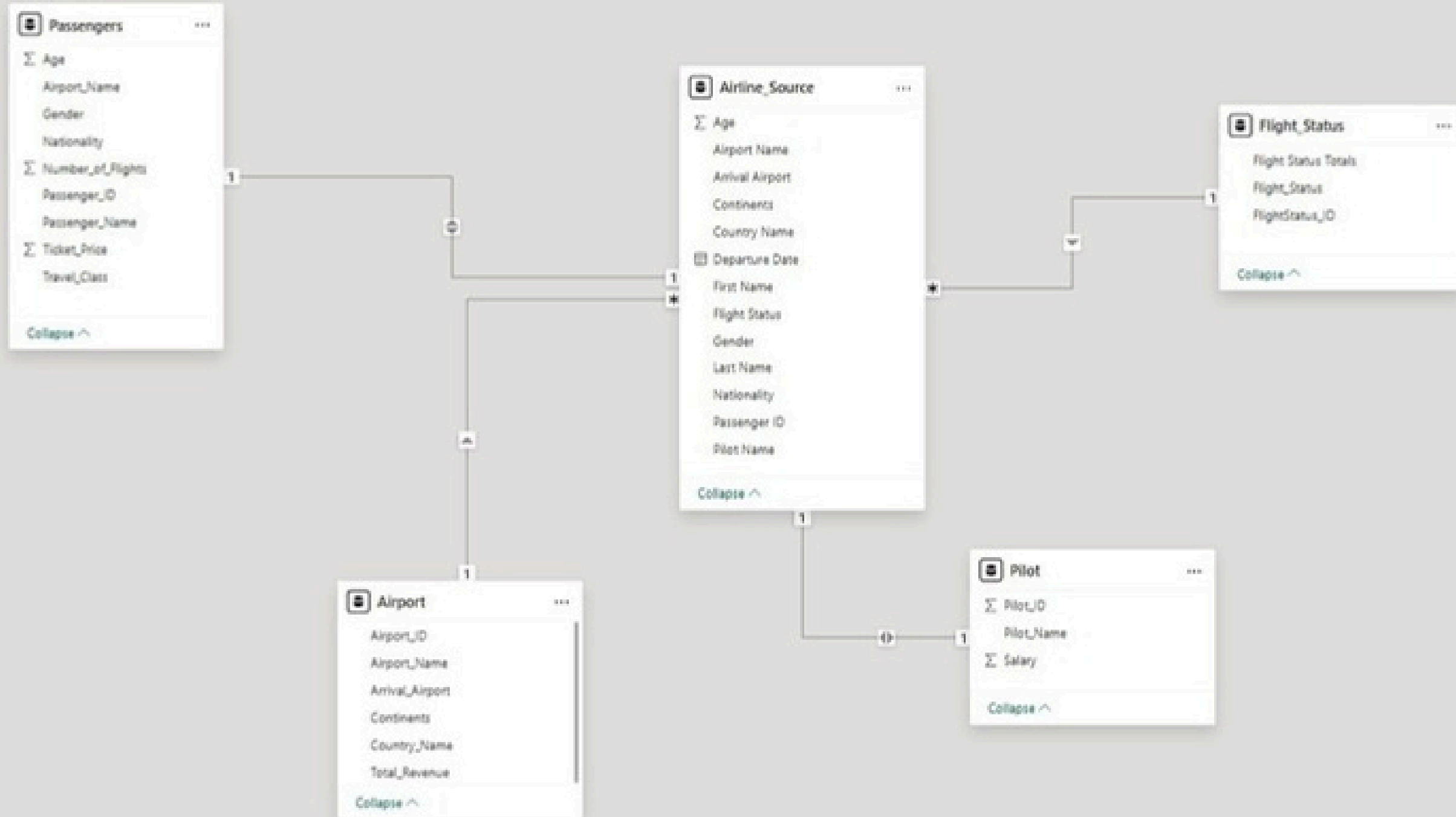
Server
127.0.0.1:3306

Database
airline

▸ Advanced options

OK Cancel

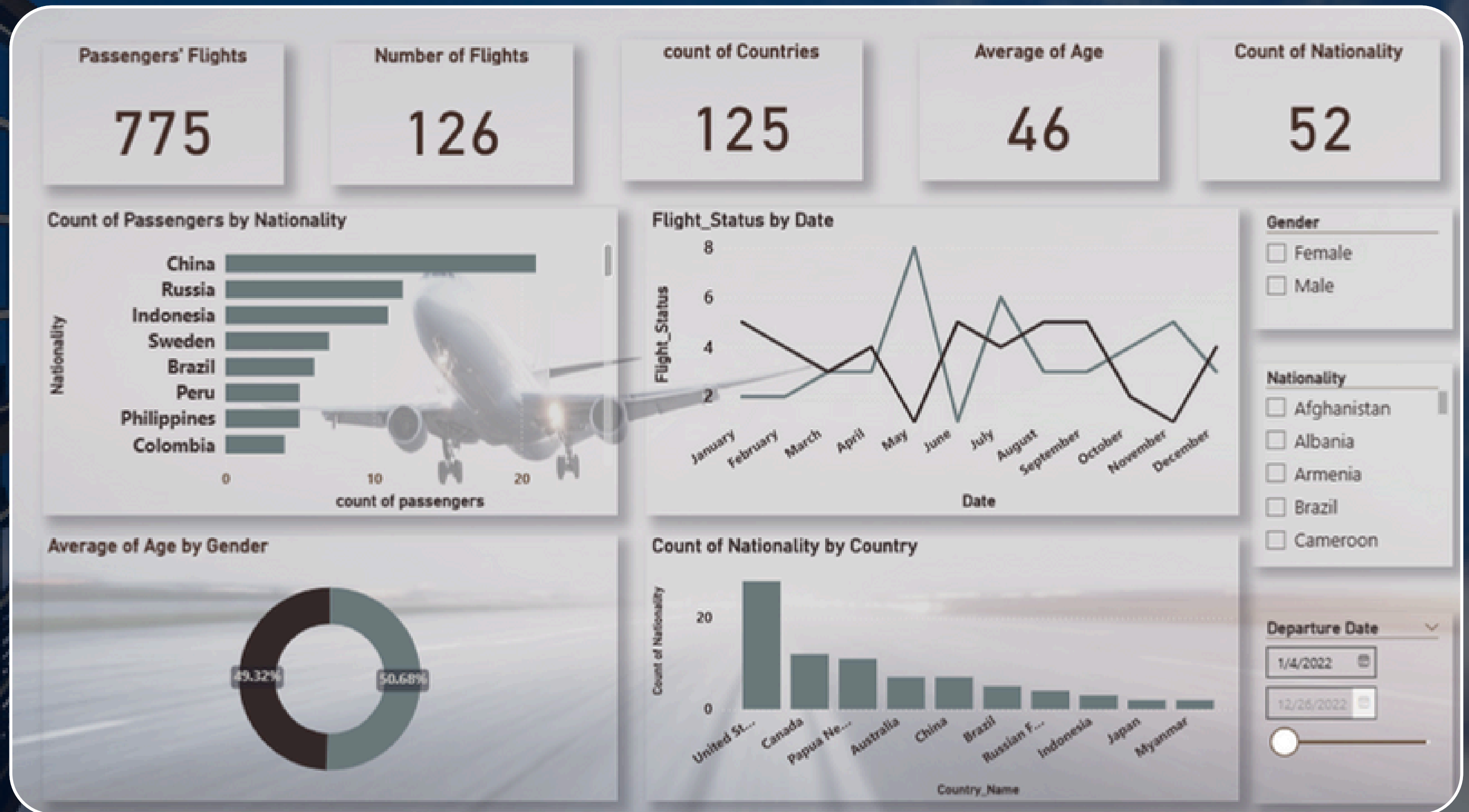
Model View



DashBoard



DashBoard



THANK YOU

